~~$Z = \{x_1, x_2, \dots, x_n\}$~~

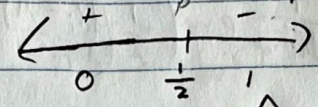
HW5 1) a) $H_0: p \le C_0$, $H_1: p > C_0$ $\quad L(z;p) = (1-p)^{z-1} \cdot p$

~~If $p \le C_0$, then $L_1 = \prod (1-p)^{x_i - 1} \cdot p$, $L_2 = \prod (1-C_0)^{x_i - 1} \cdot C_0$~~

$\text{Max } (1-p)^{z-1} \cdot p \qquad \text{Max}' = (z-1)(1-p)^{z-2} \cdot p + (1-p)^{z-1} = 0$

$(z-1)(1-p)^{-1} \cdot p + 1 = 0 \qquad (z-1)\frac{p}{1-p} + 1 = 0 \qquad \frac{p}{1-p} = \frac{-1}{(z-1)}$

$L_2: \left(1-\frac{1}{2}\right)^{z-1} \cdot \frac{1}{2}$



$\hat{p}_0 : \begin{cases} \frac{1}{2}; & \frac{1}{2} \le C_0 \\ C_0; & \text{otherwise} \end{cases}$ $\quad$ Explanation, increasing derivative

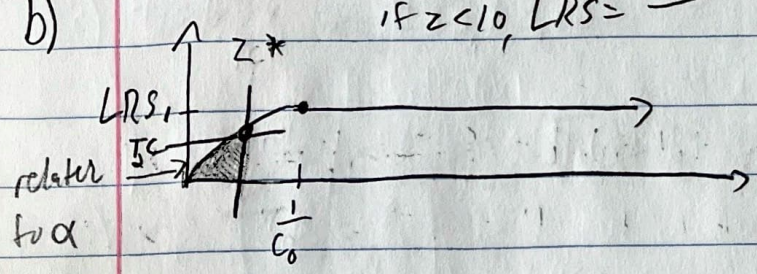
This is fixed

$L_1: (1-\hat{p}_0)^{z-1} \cdot \hat{p}_0 \qquad LRS: \frac{(1-\hat{p}_0)^{z-1} \cdot \hat{p}_0}{\left(1-\frac{1}{2}\right)^{z-1} \cdot \frac{1}{2}} \rightarrow \begin{cases} 1; & \frac{1}{2} \le C_0 \\ \frac{(1-C_0)^{z-1} \cdot C_0}{\left(1-\frac{1}{2}\right)^{z-1} \cdot \frac{1}{2}}; & \text{otherwise} \end{cases}$

Ex. $C_0 = \frac{1}{10}$ $\quad$ if $z > 10$, then $LRS = 1$

if $z < 10$, $LRS =$

b)



α in distribution of z

If we want to find TC_α, plug in z^* to LRS

$\{LR(z) < TC\} \quad = \int_0^{z^*} \frac{(1-C_0)^{z-1} \cdot C_0}{\left(1-\frac{1}{z^*}\right)^{z^*-1} \cdot \frac{1}{z^*}}$

$\Leftrightarrow_p \{z < z^*\}$

$\rightarrow \alpha$

z^* is alpha percentile of z

$z^* = \text{Geom}^{-1}_{\hat{p}_0}(\alpha)$

Poisson: $\frac{\lambda^k e^{-\lambda}}{k!}$

2) For observed sample, if it's smaller than some number, reject. This completely at α
"If all $Z_1 \ldots Z_n$ satisfy some condition, then rejection is α"
Ex: if $\lambda = 3$, our $Z = 1,2,3,4$, Pois. = .77
If $L_c \le Z_1 \ldots Z_n \le U_c$, then rejection is α.

a) For a given λ, there is a number where beyond that k value, the combined probability of all integer $k < \alpha$. We can use the pois$^{-1}(\alpha)$ given a λ to find this number. We need to use pois$^{-1}(\alpha/2)$ to find the lower end of this limit and pois$^{-1}(1-\alpha/2)$ to find the upper end. For all $Z_1 \ldots Z_n$ in this range, the probability of all Z_s will be $> 1-\alpha$. Outside this range, our probability goes down into α rejection of null territory. Since we consider the alternative hypothesis $\lambda = c$, where $c_0 < c_1$, we can remove the lower bound and just use pois$^{-1}(1-\alpha)$ to find the upper limit of our max bundry for Z_n.

b) pois$^{-1}(.95, 2) = 5, 0$. If our value Z is <0 or >5, there is only a 5% chance this will happen. pois$^{-1}(.95, 3) = 6, 1$ if $1 < Z < 6$, there is a 95% chance this will happen, meaning outside this region there is only a 5% chance, and we would reject our null.

$SS = \sum (y_i - \bar{y})^2$

3

a)

	df	SS	F	Pr(>F)
x_1	1	M1 − M2 = 2,521.5	①	.5859
x_2	1	28,478	②	.0575
Resid	n−3	96,819.4	NA	NA

① : $\frac{2521.5}{181544.5/(24-2)} = .3056$

② $\frac{28478}{155638/(24-2)} = 4.0184$

TSS = intercept model = 184,066

$RSS_{full} = 87,246.89$

$SS_{Regression} = TSS - RSS_{full} = 96,819.11$

b)

	df	SS	RSS
none	21	Nothing dropped	87246.89
X_1	1	①	155638
X_2	1	②	181544.5

① = M3 − M4 > 155638 − 87246.89 = 68391.11

② = M2 − M4 = 181544.5 − 87246.89 = 94297.61

c) $H_0: \beta_1 = \beta_2 = 0$ $H_a: \beta_1 \neq \beta_2 \neq 0$

Compare null to full model

$F = \frac{96819.11}{2} \Big/ \frac{87246.89}{21} \approx 11.652$

For $\alpha = .05$, $F(2, 21) = 3.467$. Since our F value is larger than the critical point from the F table, we reject the null in favor of the alternative. There is significant evidence to show that the X_1 or X_2 terms are needed because $\beta_1 \neq \beta_2 \neq 0$. At least one of the terms is significant

4) Minority, Sex, and SES all seem to be statistically significant to MathAch scores, so our Model should certainly include them. SES is quantitative, while minority & sex are fixed level binary factors, so we can consider an interaction term between minority & sex easily.

$$\text{MathAch}_{ij} =$$

$$\text{LVL 1}: \beta_{0j} + \beta_{1j}(sex_{ij}) + \beta_{2j}(minority_{ij}) + \beta_{3j}(SES_{ij}) + \beta_{4j}(SES_{ij} - \overline{SES_j}) + R_{ij}$$

$$+ \beta_{5j}(sex_minority_{ij}) \quad \rightarrow R_{ij} \sim N(0, \sigma^2)$$

$$\text{LVL 2}: \beta_{0j} = \gamma_{00} + \gamma_{10}(\overline{SES_j}) + U_{0j} + U_{ij}$$

$$\beta_{1j} = \gamma_{01};\ \beta_{2j} = \gamma_{02};\ \beta_{3j} = \gamma_{03},\ \beta_{4j} = \gamma_{04},\ \beta_{5j} = \gamma_{05} \quad \rightarrow \begin{pmatrix} U_{0j} \\ U_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{01} & \tau_1^2 \end{pmatrix}\right)$$

This gives us an LMM of the form

$$\text{MathAch} = \gamma_{00} + \gamma_{10}(\overline{SES_j}) + U_{0j} + \gamma_{01}(sex_{ij}) + \gamma_{02}(minority_{ij}) + \gamma_{03}(SES_{ij}) +$$

$$\gamma_{04}(SES_{ij} - \overline{SES_j}) + \gamma_{05}(sex_minority_{ij}) + U_{1j}(\overline{SES_j} - \overline{SES_j}) + R_{ij}$$

↑ Interaction dummy term for sex & minority crosses

5) a) Intensity should be a factor, there is no given evidence to suggest that intensity = 3 is 3x more than intensity = 1, they are simply 4 different intensity levels.

Since our sd goes up significantly per intensity level, we certainly should consider it as an effect in the final model (evidence C). Also from control to dry (evidence B) the means differ by visually what looks like a lot (evidence A as well), so we should add this. The slope sd of dry & control is very similar, and by (evidence B) our all means differ significantly, so the interaction term would be helpful & could be proven later, if found to not be significant.

b) Random intercept: Yes, each subject differs in control & in dry, differing across intensity levels.

X Random dry: No, since control & dry sd are very similar compared to each other, there's no need for this random term, enough randomness would be captured by the random intercept (evidence E)

Random intensity: Because our variance increases by up to 1.92 over our intensity levels, (evidence F), it certainly would be wise to include the random intensity slope term.

c) For R_{ij} matrix, we should use a matrix to capture variances between different treatments and intensities (since var ↑ as intensity ↑), but one that has similar correlations per subject. In general it seems like an unstructured matrix works best to let the model learn all this, but we can probably add some minor restrictions to it.

d) model = lme(Response Amplitude ~ Dry + Intensity + interaction(Dry, Intensity), random ~ Intensity, data = dat, correlation = corSymm(form = ~1 | Subject))

e) $R_{ij} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} & \sigma_{24} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 & \sigma_{34} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_4^2 \end{bmatrix}$ ← 4 levels of treatment, Dry & control sd are similar so no need to separate

$G \begin{bmatrix} \tau_1^2 & \tau_{12} \\ \tau_{21} & \tau_2^2 \end{bmatrix}$ ← Only consider baseline intercept (per rat) and the variance of slope. The τ_{12} & τ_{21} are the covariance of rat & intensity, meaning some rats might react to different intensities w/ diff variance.

I

RSE = 3.2, so $\sigma^2 = 10.24$

$R_{11} = \begin{bmatrix} 10.24 & & & \\ & 10.957 & & \\ & & 14.950 & \\ & & & 19.661 \end{bmatrix}$ ← These are VARIANCES

See next page

$3.2^2 =$

$R_{11} =$	10.24	$.62 \times \sqrt{10.24} \times \sqrt{10.957}$ $= 6.567$	$.62 \times \sqrt{10.24} \cdot \sqrt{14.95}$ $= 7.671$	$.62 \times \sqrt{10.24} \times \sqrt{19.661}$ $= 8.797$
		$3.2^2 \cdot 1.07 =$ 10.957	$.62 \times \sqrt{10.957} \times \sqrt{14.95}$ $= 7.935$	$.62 \times \sqrt{10.957} \times \sqrt{19.661}$ $= 9.099$
			$3.2^2 \cdot 1.46 =$ 14.95	$.62 \times \sqrt{14.95} \times \sqrt{19.661}$ $= 10.680$
				$3.2^2 \times 1.92 =$ 19.661